Integrated Communication Circuits

Lecture 1: High Frequency Tuned Amplifier for Small Signal Amplification

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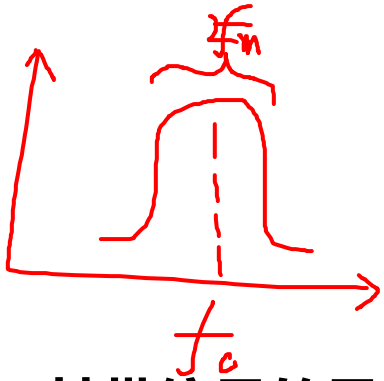
§1.1 概述:

小信号调谐放大器是一种窄带的选频放大器,是无线电接受设备的主要部件.

窄带:



$$\frac{2F_m}{f_c} << 1$$

$F_m \rightarrow$ **基带信号的最大频率**

$f_c \rightarrow$ **载波频率**

性能要求(指标):

1、增益： $A_V, A_p (A_{V0}, A_{p0})$

1-1.exe

2、通频带： BW (或 $BW_{0.707}$) $\frac{1}{\sqrt{2}}$ 3dB

3、选择性： $S = A_V / A_{V0}$

矩形系数 $K_{0.1} = BW_{0.1} / BW$

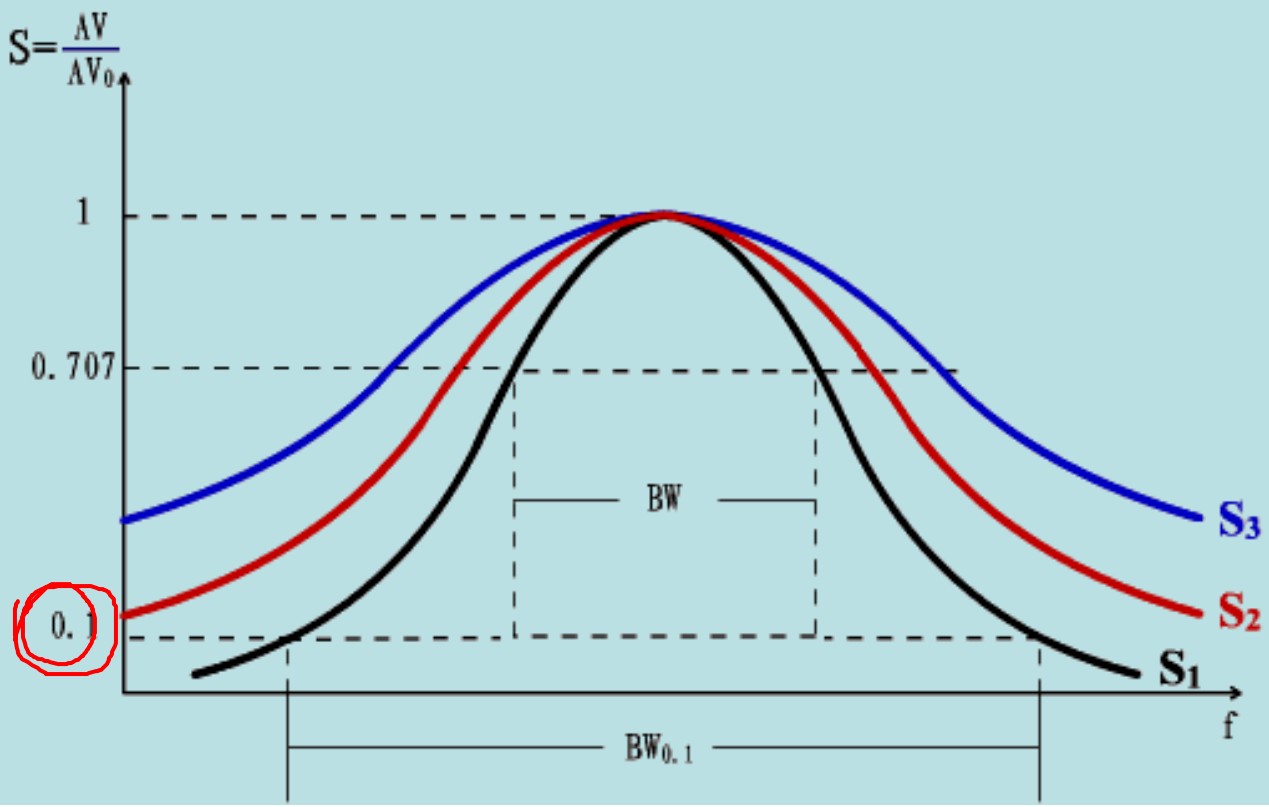
4、稳定性：

附加相移

相对增益

$$S = A_V / A_{V0}$$

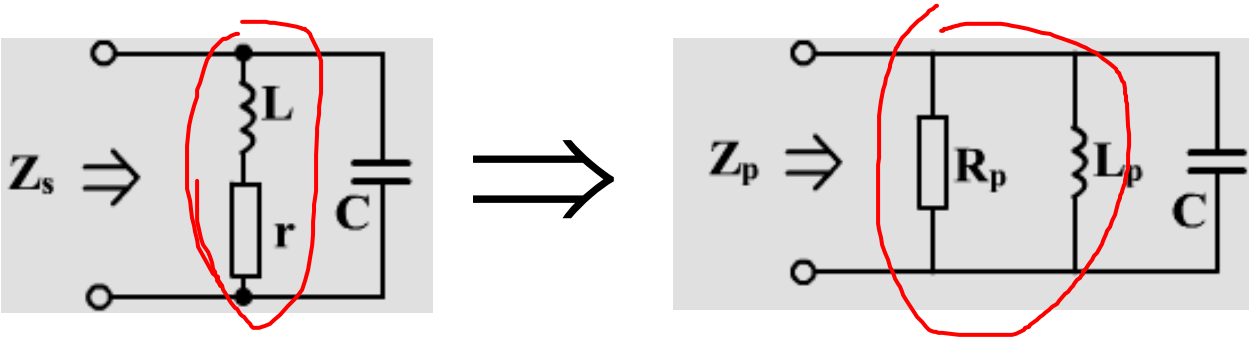
$$K_{0.1} = BW_{0.1} / BW$$ 矩形系数



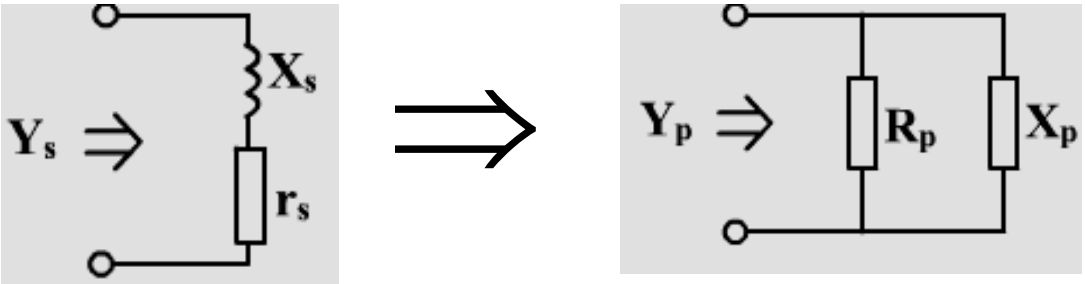
§1.2 谐振回路 Resonance

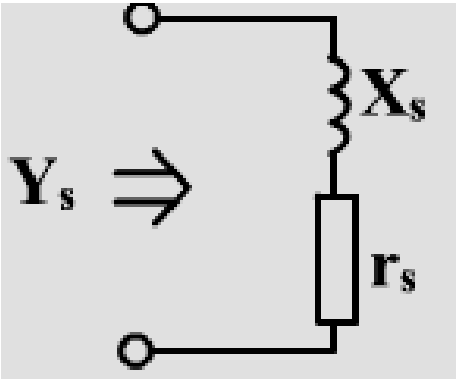
一、并联谐振回路 Parallel LC Circuits

1、串、并联转换

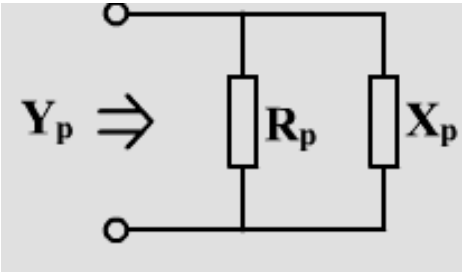


其中:





串： $Y_S = \dfrac{1}{Z_S} = \dfrac{1}{r_s + jX_s} = \dfrac{r_s}{r_s^2 + X_s^2} - j\dfrac{X_s}{r_s^2 + X_s^2}$



并： $Y_p = \dfrac{1}{Z_p} = \dfrac{1}{R_p} + \dfrac{1}{jX_p}$

$$Y_s = Y_p$$

由于： $Y_s = Y_p$

则有： $R_p = \dfrac{r_s^{\ 2} + X_s^{\ 2}}{r_s} = r_s(1+Q^2)$

$$X_p = \frac{r_s^{\ 2} + X_s^{\ 2}}{X_s} = X_s(1+\frac{1}{Q^2})$$

品质因数：$Q = \dfrac{X_s}{r_s} = \dfrac{R_p}{X_p}$

当$Q >> 1$时，

$$R_p \approx \frac{X_s^{\ 2}}{r_s} = Q^2 r_s = \frac{L}{Cr}$$

$$X_p \approx X_s, L_p \approx L$$

10

分立

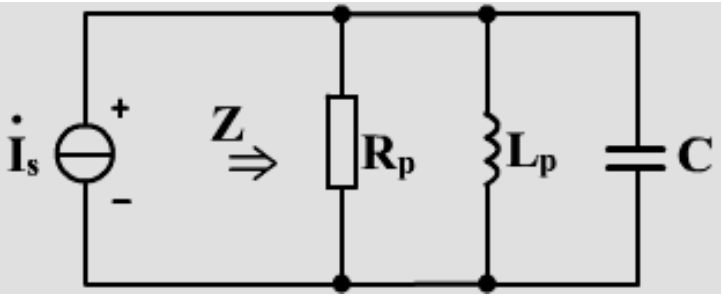
L,C

Q→100

集成

L,C

2、并联谐振回路的阻抗、谐振频率和品质因数



$$Z=\frac{1}{\frac{1}{R_p}+j(\omega C-\frac{1}{\omega L_p})}$$

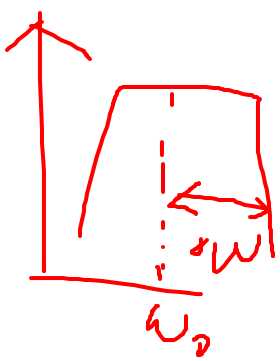
$$=\frac{R_p}{1+j(\omega CR_p-\frac{R_p}{\omega L_p})}$$

$$=\frac{R_p}{1+j(\frac{\omega}{\omega_0}-\frac{\omega_0}{\omega})Q}=\frac{R_p}{1+j\xi}$$

设：失谐系数$\xi = \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) Q$

其中$\omega_0 = \frac{1}{\sqrt{LC}}$　　$Q = R_p / \omega_0 L = R_p \omega_0 C$

当$\omega \approx \omega_0$时，$\xi = \frac{\omega^2 - \omega_0^2}{\omega\omega_0} Q$



$$= \frac{(\omega + \omega_0)\cdot(\omega - \omega_0)}{\omega_0 \cdot \omega} Q$$

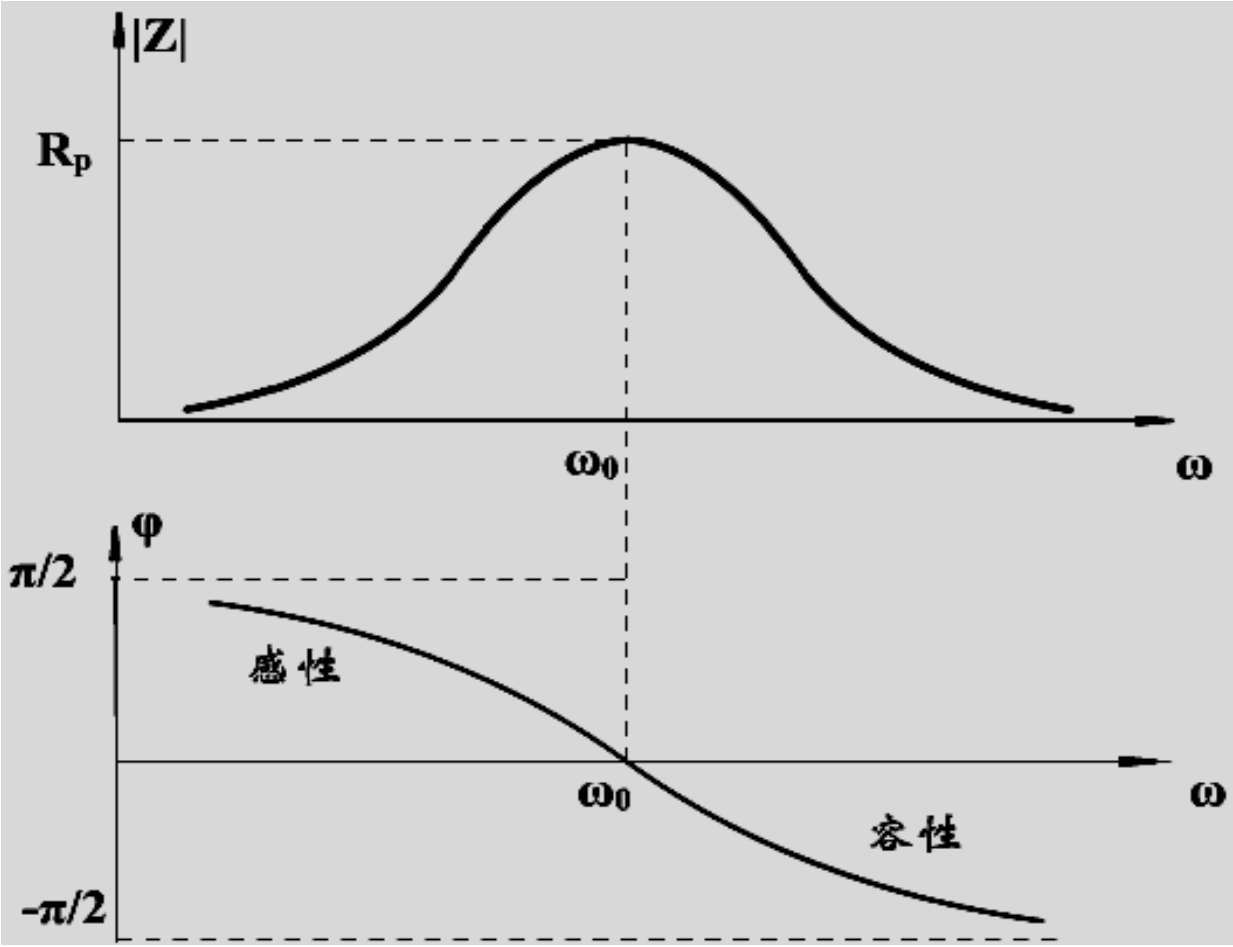
$$\approx \frac{2(\omega - \omega_0)}{\omega_0} Q$$

$$= \frac{2\Delta\omega}{\omega_0} Q$$

$$\therefore Z = \frac{R_p}{1 + j\xi} \boxed{\approx} \frac{R_p}{1 + jQ\dfrac{2\Delta\omega}{\omega_0}}$$

$$\text{幅频特性}: |Z| = \frac{R_p}{\sqrt{1+\xi^2}} = \frac{R_p}{\sqrt{1+(Q\dfrac{2\Delta\omega}{\omega_0})^2}}$$

相位特性： $\varphi = -\mathrm{arctg}\xi$


|Z|
Rp
ω0
ω
φ
π/2
感性
ω0
ω
容性
-π/2

3、 并联谐振回路的谐振曲线、通频带和选择性 S 或$K_{0.1}$

$$\because A_V = \frac{R_p}{1+j\xi}, \qquad A_{V0} = R_p$$

$$\therefore S = \frac{|A_V|}{|A_{V0}|} = \frac{1}{\sqrt{1+\xi^2}} = \frac{1}{\sqrt{1+(Q\frac{2\Delta\omega}{\omega_0})^2}}$$

$$\varphi = -\mathrm{arctg}\xi$$

$$\text{当} S=\frac{1}{\sqrt{2}}=\frac{1}{\sqrt{1+\xi^2}}\text{时}, \xi=\pm 1,$$

$$\Rightarrow \xi=\overset{BW}{\boxed{2\Delta\omega}}\frac{Q}{\omega_0}\Rightarrow BW=f_0/Q$$

$$Q=\frac{f_0}{BW}$$

$$\text{当} S=0.1=\frac{1}{\sqrt{1+\xi^2}}\text{时}, \quad \xi=\pm\sqrt{99},$$

$$\Rightarrow \xi=\boxed{2\Delta\omega}\frac{Q}{\omega_0}, \Rightarrow BW_{0.1}=\sqrt{99}\frac{f_0}{Q}=9.95\frac{f_0}{Q}$$

$$\therefore K_{0.1}=\frac{BW_{0.1}}{BW}=9.95$$

4、复频域上分析

$$Z(s)=Z(j\omega)\Big|_{j\omega=s}=\frac{1}{\frac{1}{R_p}+SC+\frac{1}{SL_p}}$$

数值

物理

$$=\frac{1}{C}\cdot\frac{S}{S^2+\frac{1}{R_pC}S+\frac{1}{L_pC}}$$

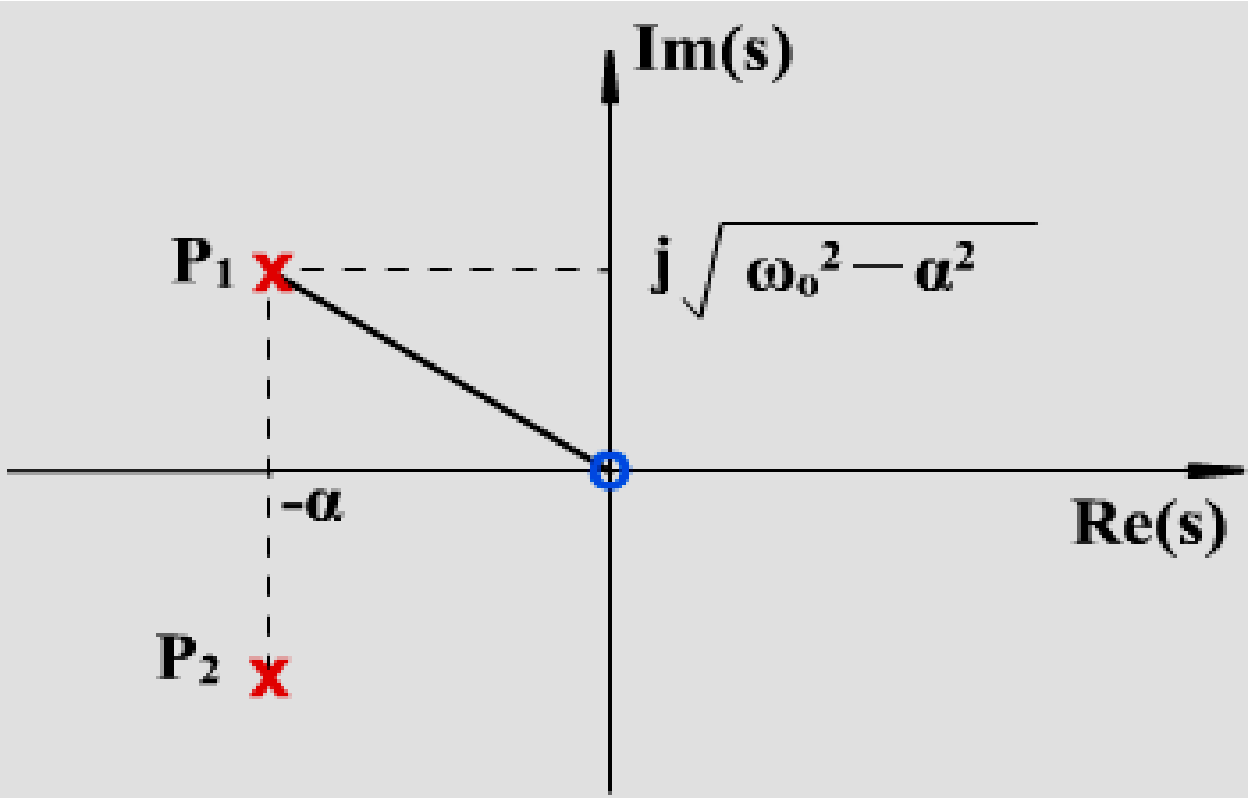
$$=\frac{1}{C}\cdot\frac{S}{S^2+2\alpha S+\omega_0^{\ 2}}=\frac{1}{C}\cdot\frac{S}{(S-p_1)\cdot(S-p_2)}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}, \text{回路衰减系数} \alpha = \frac{1}{2R_pC}$$

$$p_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -\alpha + \mathrm{j}\sqrt{\omega_0^2 - \alpha^2}$$

$$p_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -\alpha - \mathrm{j}\sqrt{\omega_0^2 - \alpha^2}$$

$$BW = \frac{f_0}{Q} = \frac{1}{2\pi} \times \frac{\omega_0}{Q} = \frac{1}{2\pi} \times \frac{1}{CR_p} = \boxed{\frac{2\alpha}{2\pi}} = 2 \cdot \frac{\alpha}{2\pi}$$



α代表通频带的一半 带宽

例1

一并联LC电路，已知电感为3mH，绕线阻抗2Ω，电容为0.47μF。试求：

(1)f_0；(2)Q；(3)R_p；(4)BW；(5)$K_{0.1}$；(6)A_{v0}；

$$Q = \frac{R_p}{\omega_0 L} = R_p \cdot \omega_0 C$$

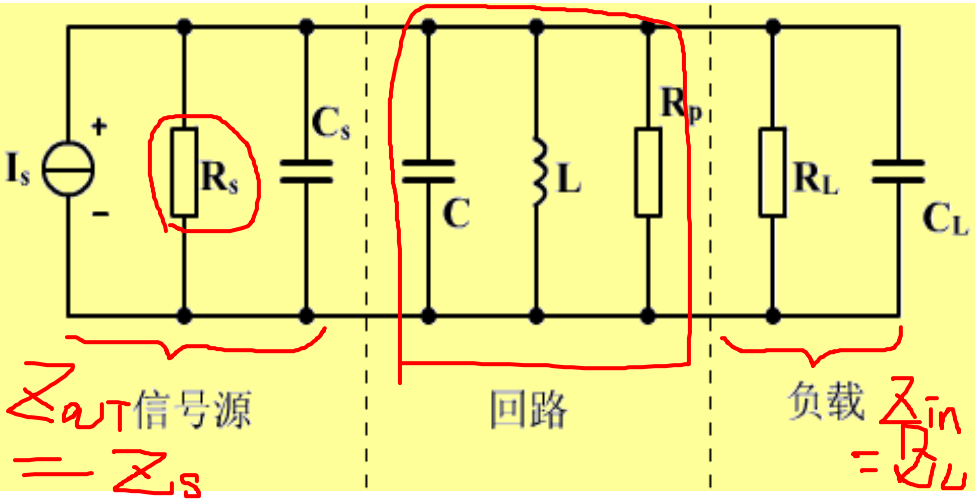
解

$$\text{BW} = \frac{f_0}{Q} = \frac{\omega_0 L}{r_s} \qquad r_s \approx \frac{R_p}{Q^2}$$

二、阻抗变换

级联

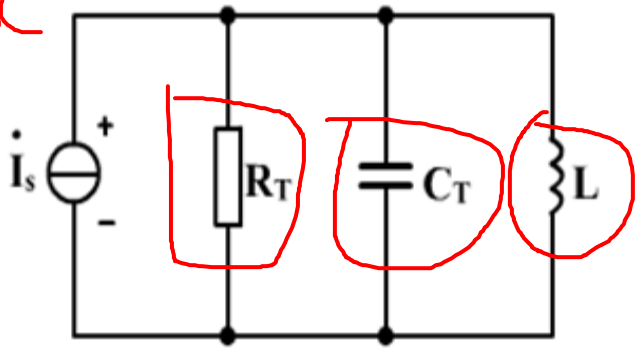
信号源内阻及负载对谐振回路的影响。



图(a)

$Q = R_p \omega_0 C$ → unloaded Q

等效 →



$Q_L = R_T / \omega_0 L$

图(b)

（b）图中：$R_T = R_S // R_P // R_L \downarrow Q_L = R_T / \omega_0 L \downarrow . < Q_0$

$$C_T = C_S + C + C_L \uparrow \omega_0' = \frac{1}{\sqrt{LC_T}} \downarrow . < \omega_0$$

谐振回路的前一级（或信号源）

下一级（或负载）的阻抗

对谐振回路的性能影响较大

增益下降

选择性变差

谐振频率漂移

解决这一问题的方法之一：

采用部分接入法，

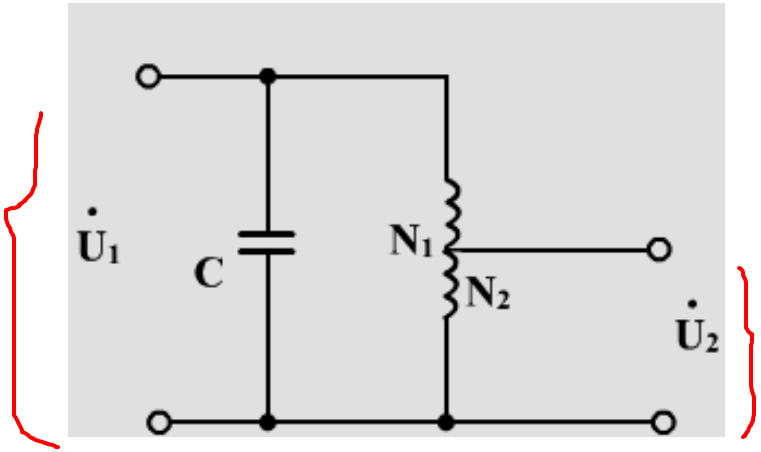
减小谐振回路的前一级（或信号源）和下一级（或负载）的阻抗对谐振回路的影响。

部分接入的电路：

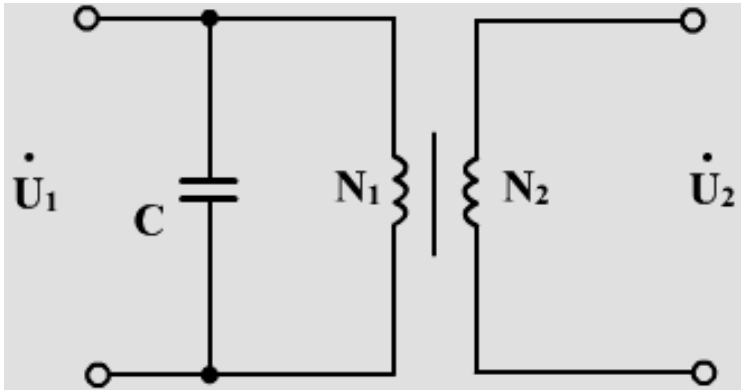
- 变压器耦合
- 电感分压
- 电容分压

(1)接入系数 p

定义为接入电压U_2与回路总电压U_1的比值

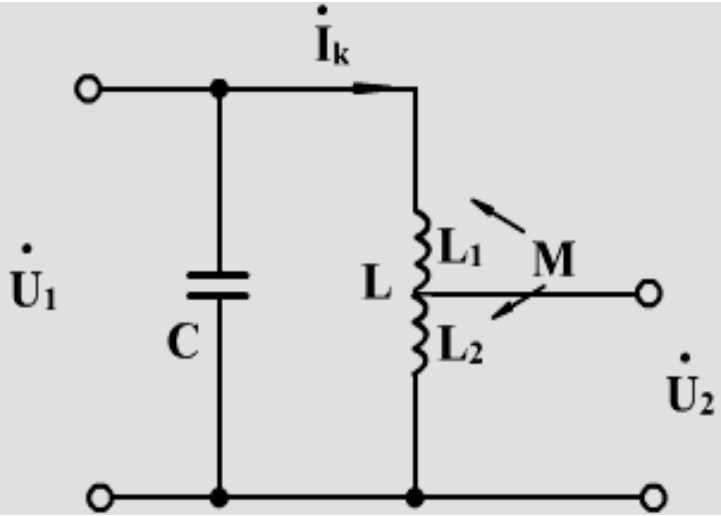


图(a)自耦变压器.



图(b)变压器.

图(a)(b)的接入系数：$p = \dfrac{N_2}{N_1}$



图(c)：

$$p=\frac{\dot{U}_2}{\dot{U}_1}=\frac{\dot{I}_k j\omega(L_2\pm M)}{\dot{I}_k j\omega(L_1+L_2\pm M)}$$

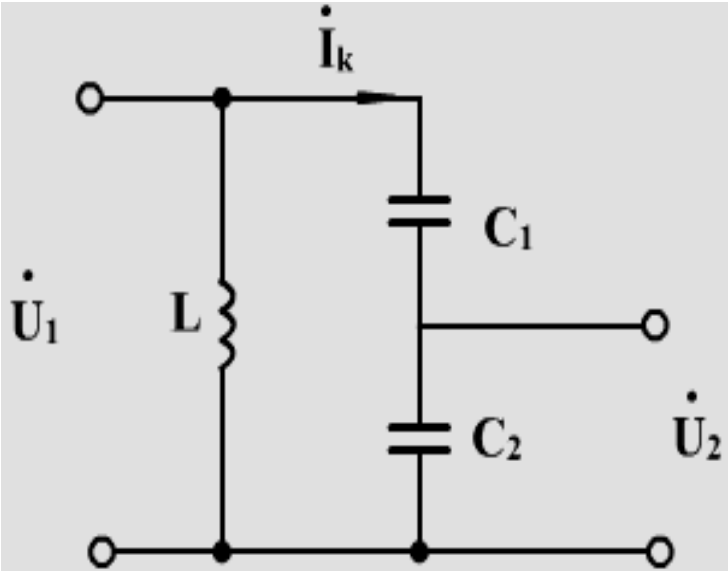
$$=\frac{L_2\pm M}{L}$$

图(c)电感分压

$$总电感：L=L_1+L_2\pm 2M.$$

$$若M=0 \qquad p=\frac{L_2}{L}=\frac{L_2}{L_1+L_2}$$

图(d)



图(d)电容分压

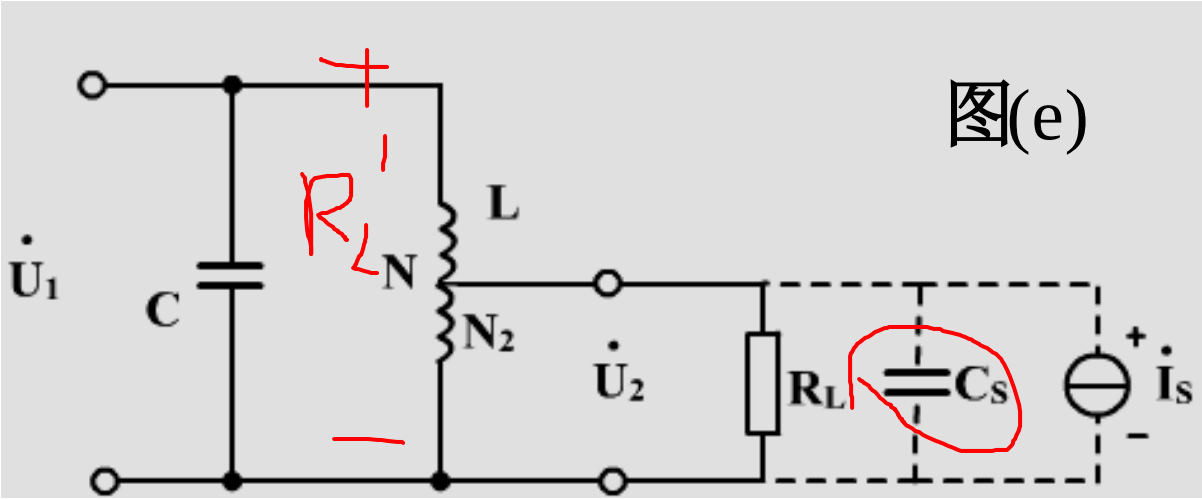
$$p=\frac{\dot{U}_2}{\dot{U}_1}=\frac{\dot{I}_k\dfrac{1}{j\omega C_2}}{\dot{I}_k\dfrac{1}{j\omega C}}$$

$$=\frac{C}{C_2}$$

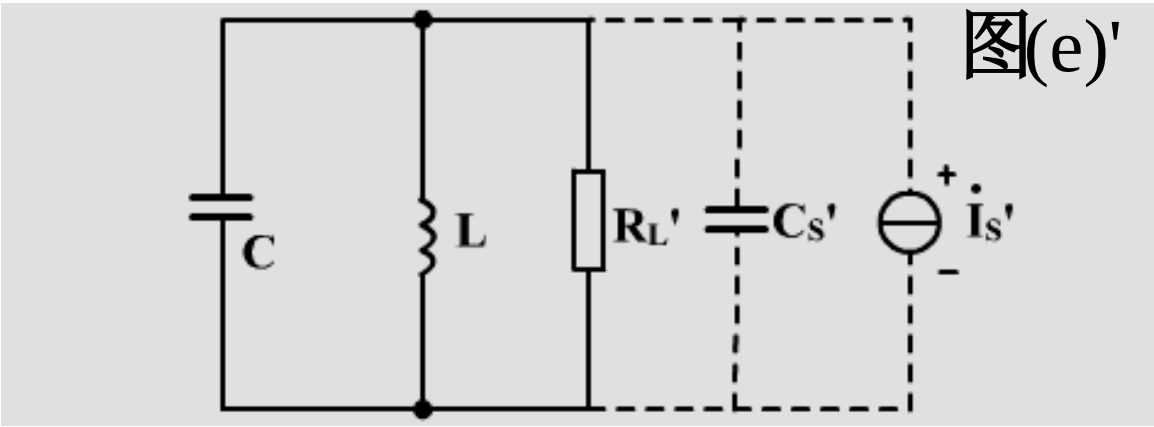
$$\text{总电容}:C=\frac{C_1C_2}{C_1+C_2}$$

$$\therefore p=\frac{C_1}{C_1+C_2}$$

(2)负载阻抗的部分接入:



R_T



(e)图R_L吸收功率：$\dfrac{U_2^{\ 2}}{R_L}$

(e)'图R_L'吸收功率：$\dfrac{U_1^{\ 2}}{R_L'}$

$R_L' \gg R_L$

两者相等：$\dfrac{U_2^{\ 2}}{R_L} = \dfrac{U_1^{\ 2}}{R_L'}$

$$\therefore R_L' = \left(\frac{U_1}{U_2}\right)^2 R_L = \frac{R_L}{p^2}$$

$p<1$

同理：$C_s' = p^2 C_s, \quad \dot{I}_s' = p\dot{I}_s$

例2

(1)在调谐放大器的负载回路两端并联电阻，试问对放大器的谐振增益、通 频带、选择性等各有什么影响？

(2)若调谐放大器的回路电感 L 的损耗电阻增大，重复以上问题.

r_s

解

$$\text{并联谐振阻抗}R_T \downarrow \rightarrow A_{V0} \downarrow$$

$$Q_L = \omega_0 C R_T \downarrow \rightarrow BW \uparrow$$

$$S = \frac{1}{\sqrt{1+\xi^2}} = \frac{1}{\sqrt{1+(2Q_L\frac{\Delta\omega}{\omega_0})^2}} \uparrow$$

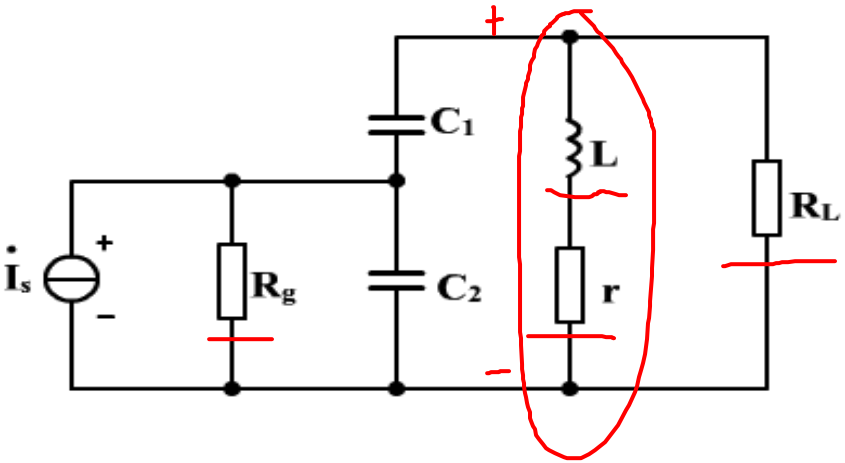
$S\uparrow$ 选择性变差， 而 $K_{0.1}=\dfrac{BW_{0.1}\uparrow}{BW\uparrow}$ 不变

（2） $r\uparrow\rightarrow R_p=\dfrac{X_s^{\ 2}}{r}\downarrow\rightarrow R_T\downarrow$ 影响同上.

$r_s\uparrow$. $Q\downarrow$ $\dfrac{\omega_0 L}{r_s}=\dfrac{R_p}{\omega_0 L}$

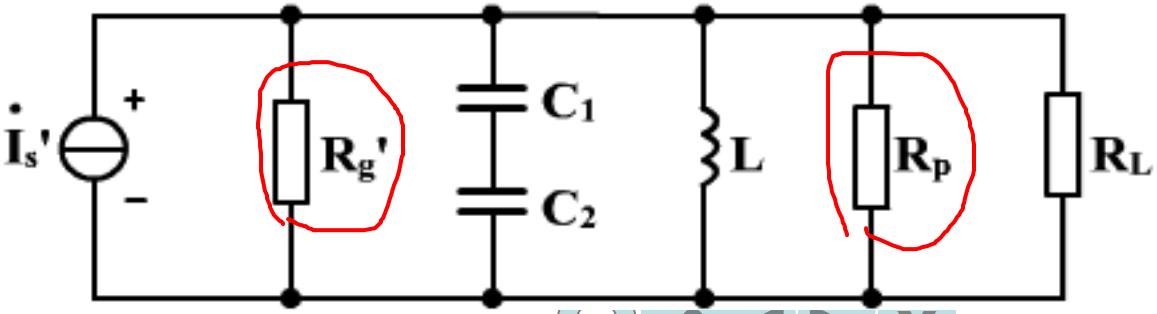
例3

题图的电容式分压并联谐振电路参数：L=200μH, r=8Ω, C_1=140pF, C_2=1400pF, R_g=5kΩ, R_L=100kΩ.
求 f_0 和 BW。



解

高频电路等效:



$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{L\dfrac{C_1C_2}{C_1+C_2}}} = \cdots = 1\text{MHz}$$

$$Q_0 = \frac{\omega_0 L}{r} = \frac{2\pi\times10^6\times200\times10^{-6}}{8} = 157$$

$$R_p = rQ_0^{\ 2} = \frac{L}{Cr} = 8\times157^2 = 197\text{k}\Omega$$

$$p = \frac{C_1}{C_1+C_2} = 0.091 \qquad R_g' = \frac{R_g}{P^2} = \frac{5\times10^3}{0.091^2} = 604\text{k}\Omega$$

$$R_{\Sigma} = R_g' // R_p // R_L = 60\text{k}\Omega$$

$$Q_L = \frac{R_{\Sigma}}{\omega_0 L} = \frac{60\times10^3}{2\pi\times10^6\times200\times10^{-6}} = 47.8$$

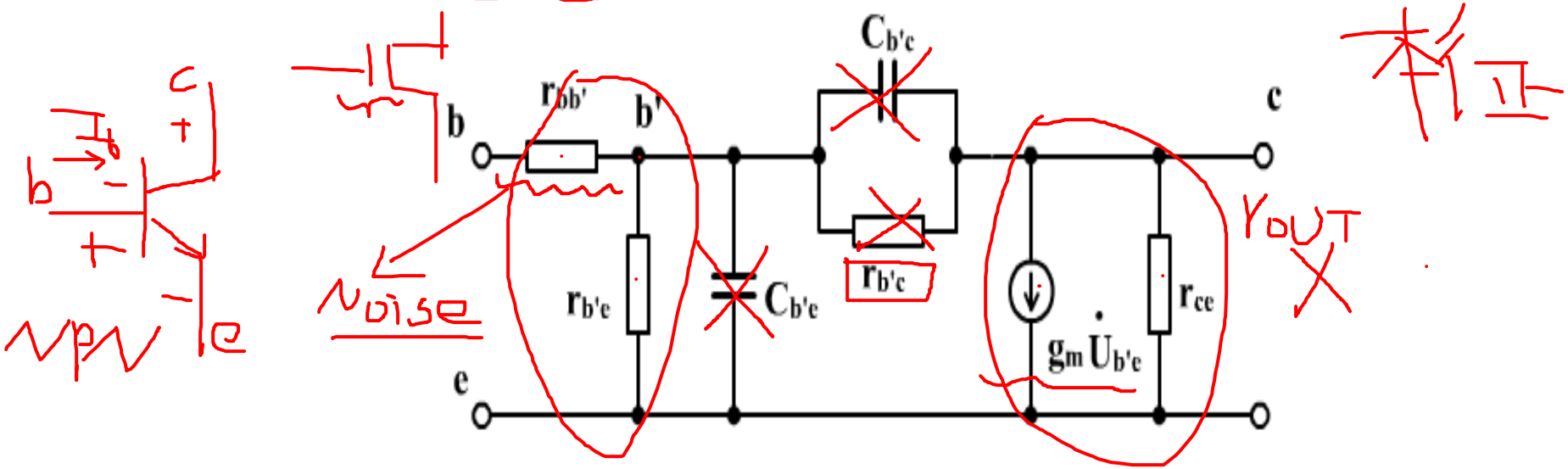
$$\therefore BW = \frac{f_0}{Q_L} = \frac{10^6}{47.8} = 20.92\text{kHz.}$$

§1.3 单级单回路调谐放大器

一、晶体管等效电路

1.混合π参数等效电路.

Bipolar ↔ CMOS



混π参数在一定频率范围内与频率无关,只与工作点有关,因此可用于宽带分析.

低频

2. Y参数等效电路：

在小信号运用时,晶体管可近似为线性网络,因此可用线性方法求解

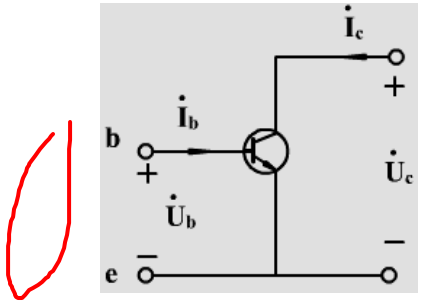
$$\boxed{\begin{aligned}\dot{I}_b &= y_{ie}\dot{U}_b + y_{re}\dot{U}_c \\ \dot{I}_c &= y_{fe}\dot{U}_b + y_{oe}\dot{U}_c\end{aligned}}$$

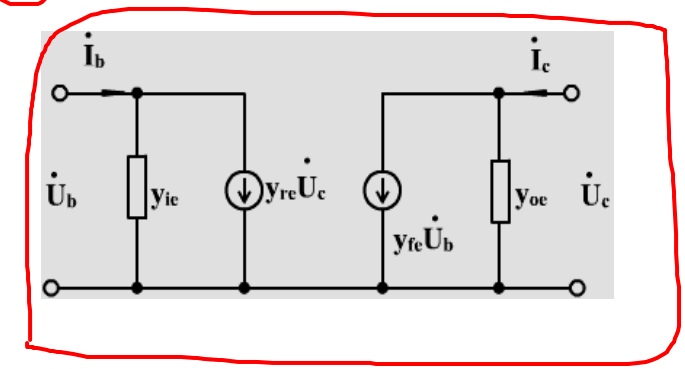
y

[ ]2x2

r = reverse

f =



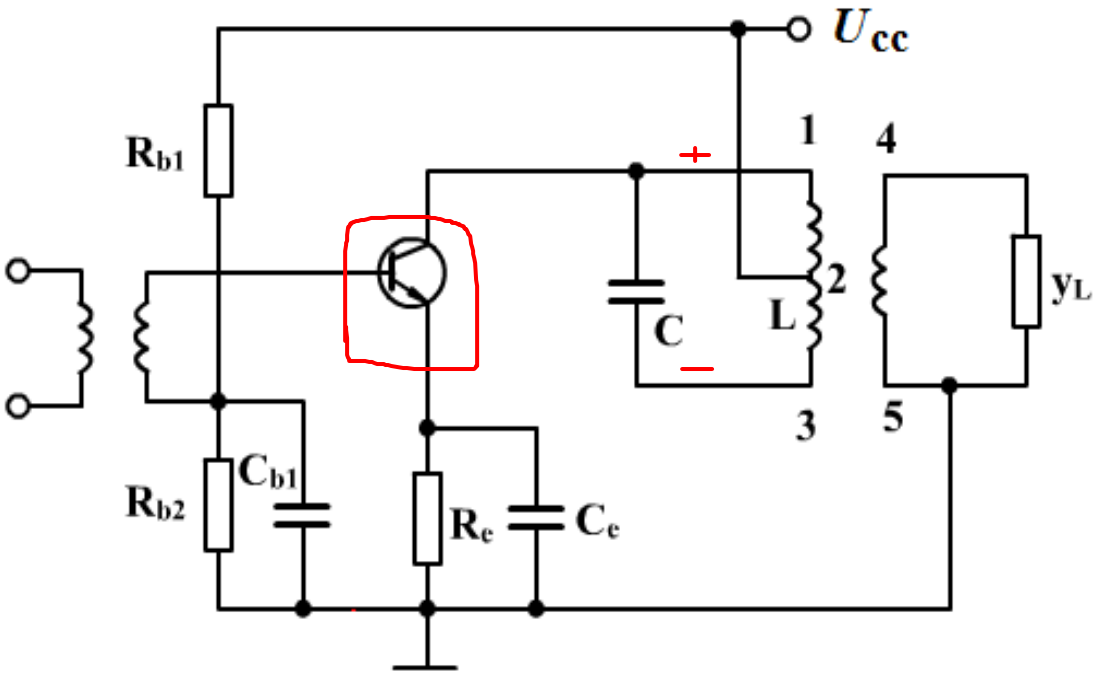


Y参数不仅与工作点有关而且与频率有关,因此只适用于窄带分析。

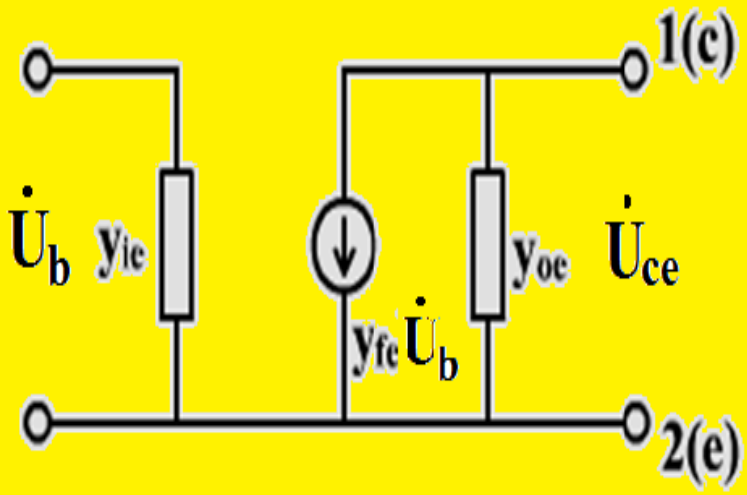
二、电路组成

对于线路要把握 “**管为路用，以路为主**” 的原则。

(a)图：



(b)图：晶体管等效为Y参数



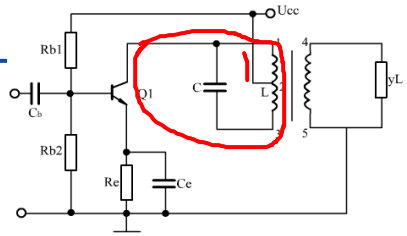
- **晶体管接入回路的接入系数：** $p_1 = \dfrac{N_{12}}{N}$
- **负载端的接入系数：** $p_2 = \dfrac{N_{45}}{N}$负载端

折算到LC回路两端:

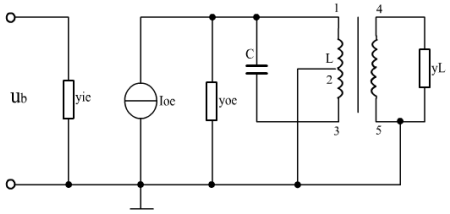
$$g_{oe}' = p_1^2 g_{oe}$$

$$C_{oe}' = p_1^2 C_{oe}$$

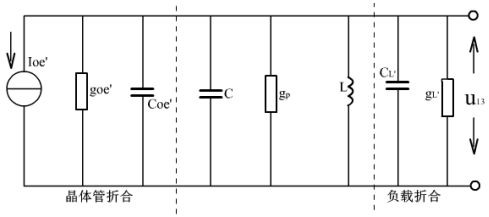
$$\dot{I}_{oe}' = p_1 \dot{I}_{oe} = p_1 y_{fe} \dot{U}_b$$



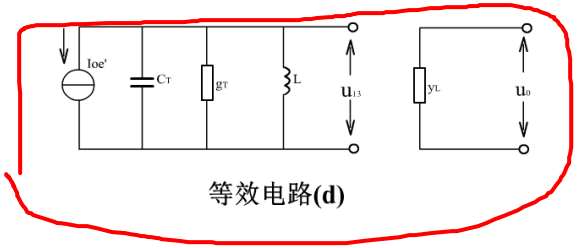
共射放大电路



等效电路(b)



等效电路(c)



等效电路(d)

同理负载端: $g_L' = p_2^2 g_L$

$$c_L' = p_2^2 c_L$$

最后合并同类元件:

$$C_T = p_1^2 C_{oe} + p_2^2 C_L + C$$

$$G_T = p_1^2 g_{oe} + p_2^2 g_L + G_p$$

$$Q_0 = \frac{1}{\omega_0 L G_p}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC_T}} = \frac{1}{2\pi\sqrt{L(P_1^2 C_{oe} + P_2^2 C_L + C)}}$$

$$Q_L = \frac{\omega_0 C_T}{G_T} = \frac{1}{\omega_0 L G_T} = \frac{1}{\omega_0 L(P_1^2 g_{oe} + P_2^2 g_L + G_p)}$$

三、计算

输出电压：

$$\dot{U}_0 = p_2\dot{U}_{13} = -\frac{p_1 p_2 |y_{fe}| \dot{U}_b}{G_T(1+j\xi)}$$

电压增益：

$$A_V = \frac{\dot{U}_0}{\dot{U}_b} = -\frac{p_1 p_2 |y_{fe}|}{G_T(1+j\xi)}$$

$$A_{V_0} = \frac{-p_1 p_2 |y_{fe}|}{G_T}$$

输出功率：$P_0 = U_0^{\,2} g_L = \left(\dfrac{p_1 p_2 |y_{fe}| U_b}{G_T} \right)^2 g_L$

输入功率：$P_i = U_b^{\,2} g_{ie}$

功率增益

$$\therefore A_{p0} = \frac{P_0}{P_i} = \frac{\dot{U}_0^{\,2}}{\dot{U}_b^{\,2}} \cdot \frac{g_L}{g_{ie}} = A_{V0}^2 \cdot \frac{g_L}{g_{ie}}$$

$$= \left(\frac{p_1 p_2 |y_{fe}|}{G_T} \right)^2 \cdot \frac{g_L}{g_{ie}}$$

当阻抗匹配时（不考虑回路本身损耗$G_p=0$）

$$p_1^2 g_{oe} = p_2^2 g_L,$$

$$G_T = p_1^2 g_{oe} + p_2^2 g_L = 2p_1^2 g_{oe}$$

$$\therefore A_{po\max} = \frac{(p_1 p_2 |y_{fe}|)^2}{4p_1^4 g_{oe}^2} \times \frac{g_L}{g_{ie}}$$

$$= \frac{p_1^2 p_2^2 |y_{fe}|^2}{4p_1^4 g_{oe}^2} \times \frac{p_1^2 g_{oe}}{p_2^2 g_{ie}} = \frac{|y_{fe}|^2}{4g_{oe} g_{ie}}$$

$$A_{po\max} = \frac{|y_{fe}|^2}{4g_{oe} g_{ie}}$$

功率增益 A_{po} 与最大功率增益 A_{pomax} 之间的关系：

$$A_{po}=\frac{|y_{fe}|^2 p_1^2 p_2^2}{G_T^2}\times\frac{g_L}{g_{ie}}\times\frac{g_{oe}}{g_{oe}}=\frac{|y_{fe}|^2 g_{oe}' g_L'}{G_T^2 g_{ie} g_{oe}}$$

$$=\frac{|y_{fe}|^2}{4g_{ie}g_{oe}}\times\frac{(g_{oe}'+g_L')^2}{G_T^2}\times\frac{4g_{oe}'g_L'}{(g_{oe}'+g_L')^2}$$

$$=A_{pomax}\times\eta_1\times\eta_2$$

其中η_1为回路插入损耗：

$$\eta_1 = \frac{(g_{oe}'+g_L')^2}{G_T^{\ 2}} = \frac{(g_{oe}'+g_L'+g_p-g_p)^2}{G_T^{\ 2}}$$

$$= (1-\frac{g_p}{G_T})^2 = (1-\frac{Q_L}{Q_0})^2$$

η_1描述了由于电感线圈损耗不能忽略带来的影响。

$$\eta_1 = (1-\frac{Q_L}{Q_0})^2 < 1$$

其中η_2为失配功率：

$$\eta_2 = \frac{4g_{oe}'g_L'}{(g_{oe}'+g_L')^2} \times \frac{g_{oe}'^2}{g_{oe}'^2} = \frac{4\frac{g_L'}{g_{oe}'}}{(1+\frac{g_L'}{g_{oe}'})^2} = \boxed{\frac{4D}{(1+D)^2}}$$

其中$D = \frac{g_L'}{g_{oe}'}$称为失配系数

$$\eta_2 = \frac{4D}{(1+D)^2} \le 1$$

阻抗匹配时，p_1、p_2最佳选取：

$$\because\ p_1^2 g_{oe} = p_2^2 g_L = \frac{1}{2}(G_T - G_p)$$

$$\therefore\ p_1^{\ 2} = \frac{(G_T - G_p)}{2g_{oe}}$$

$$p_2^{\ 2} = \frac{(G_T - G_p)}{2g_L}$$

§1.4 多级单调谐放大器

1. 电压增益和通频带

n 级放大器谐振点的总增益为

$$A_{V0\Sigma} = A_{V1} \cdot A_{V2} \cdot A_{V3} \cdots A_{Vn}$$

选择性

$$S = \frac{A_{V\Sigma}}{A_{V0\Sigma}} = \frac{1}{(1 + j2Q_L \frac{\Delta\omega}{\omega_0})^n}$$

$$单级BW = \frac{f_0}{Q_L}$$

通频带

$$BW_n = \frac{f_0}{Q_L}\sqrt{2^{1/n}-1}$$

$\sqrt{2^{1/n}-1}$为带宽缩减因子

2. 矩形系数

$$K_{0.1n} = \frac{\sqrt{100^{1/n}-1}}{\sqrt{2^{1/n}-1}}$$

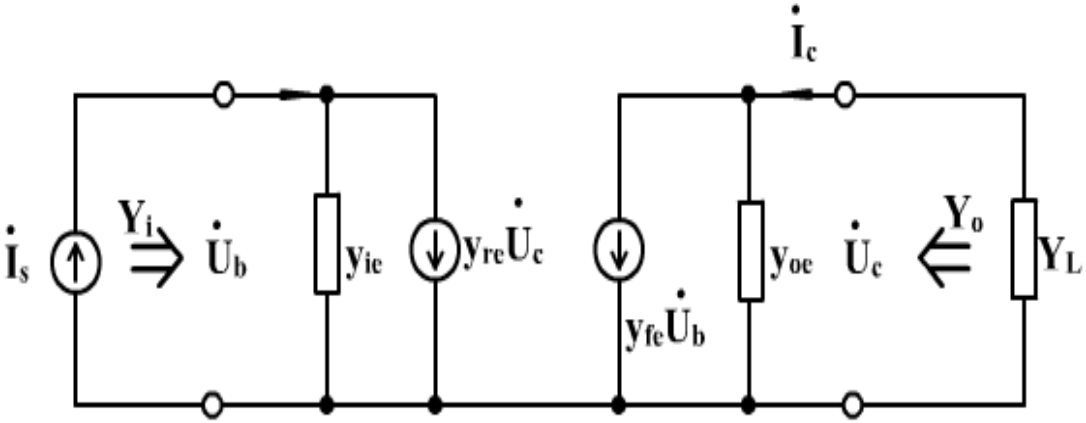
n	1	2	3	4	5	6
$\sqrt{2^{1/n}-1}$	1.00	0.64	0.51	0.43	0.39	0.35
$K_{0.1}$	9.95	4.80	3.76	3.40	3.20	3.10

可见，当n 增大时，矩形系数 $K_{0.1}$减小，

选择性得到了改善。

§1.5 放大器稳定性问题：

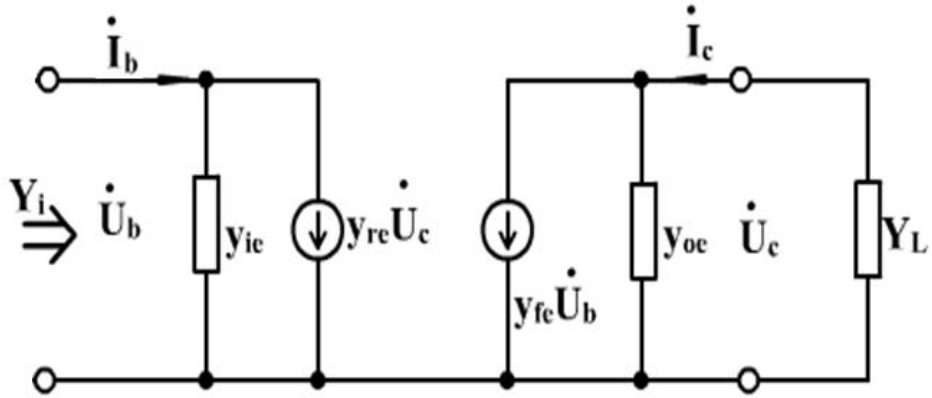
高放不稳定的原因是管子内部存在反馈,即有$C_{b'c}$或y_{re}的存在.放大器不稳定主要表现是频率特性变坏,性能指标变化,严重时会产生自激



$$Y_i = \frac{\dot{I}_b}{\dot{U}_b} = y_{ie} + \dot{Y}_i'$$

$$Y_i' = -\frac{y_{re}y_{fe}}{(y_{oe} + Y_L)}$$

$$Y_0 = y_{oe} - \frac{y_{re}y_{fe}}{y_{ie} + Y_s} = y_{oe} + Y_0'$$



$$\dot{I}_b = y_{ie}\dot{U}_b + y_{re}\dot{U}_c \qquad (1)$$

$$\dot{I}_c = y_{fe}\dot{U}_b + y_{oe}\dot{U}_c \qquad (2)$$

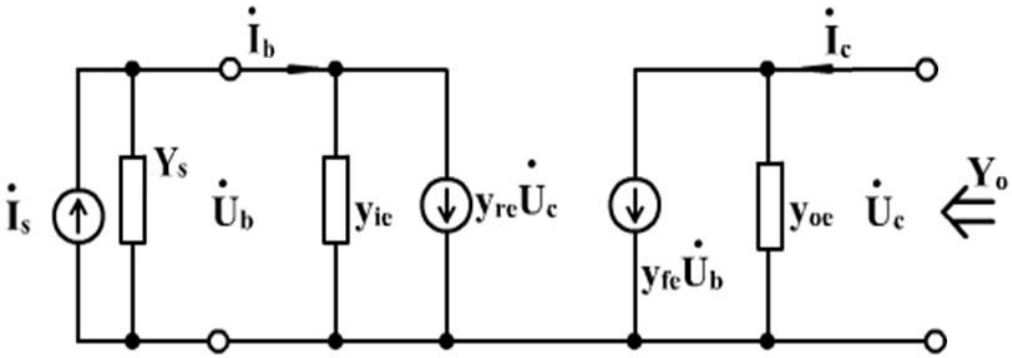
$$\dot{I}_c = -Y_L\dot{U}_c \qquad (3)$$

(3)**式代入**(2)**式，有：**

$$\dot{U}_c = -\frac{y_{fe}}{y_{oe} + Y_L}\dot{U}_b \qquad (4)$$

(4)**式代入**(1)**式，有：**

$$Y_i = \frac{\dot{I}_b}{\dot{U}_b} = y_{ie} - \frac{y_{re}y_{fe}}{y_{oe} + Y_L}$$



$$\dot{I}_b = y_{ie}\dot{U}_b + y_{re}\dot{U}_c \qquad (1)$$

$$\dot{I}_c = y_{fe}\dot{U}_b + y_{oe}\dot{U}_c \qquad (2)$$

$$\dot{I}_b = -Y_S\dot{U}_b \qquad (3)$$

(3)**式代入**(1)**式，有：**

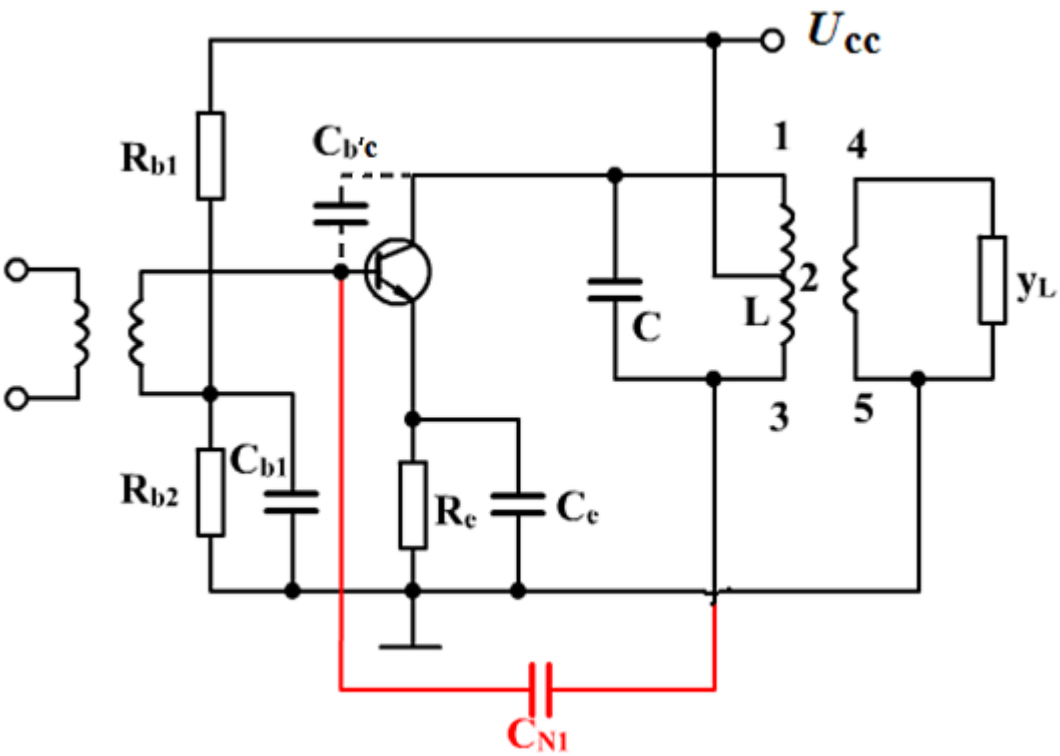
$$\dot{U}_b = -\frac{y_{re}}{y_{ie} + Y_S}\dot{U}_c \qquad (4)$$

(4)**式代入**(2)**式，有：**

$$Y_0 = \frac{\dot{I}_c}{\dot{U}_c} = y_{oe} - \frac{y_{re}y_{fe}}{y_{ie} + Y_S}$$

提高稳定性的措施

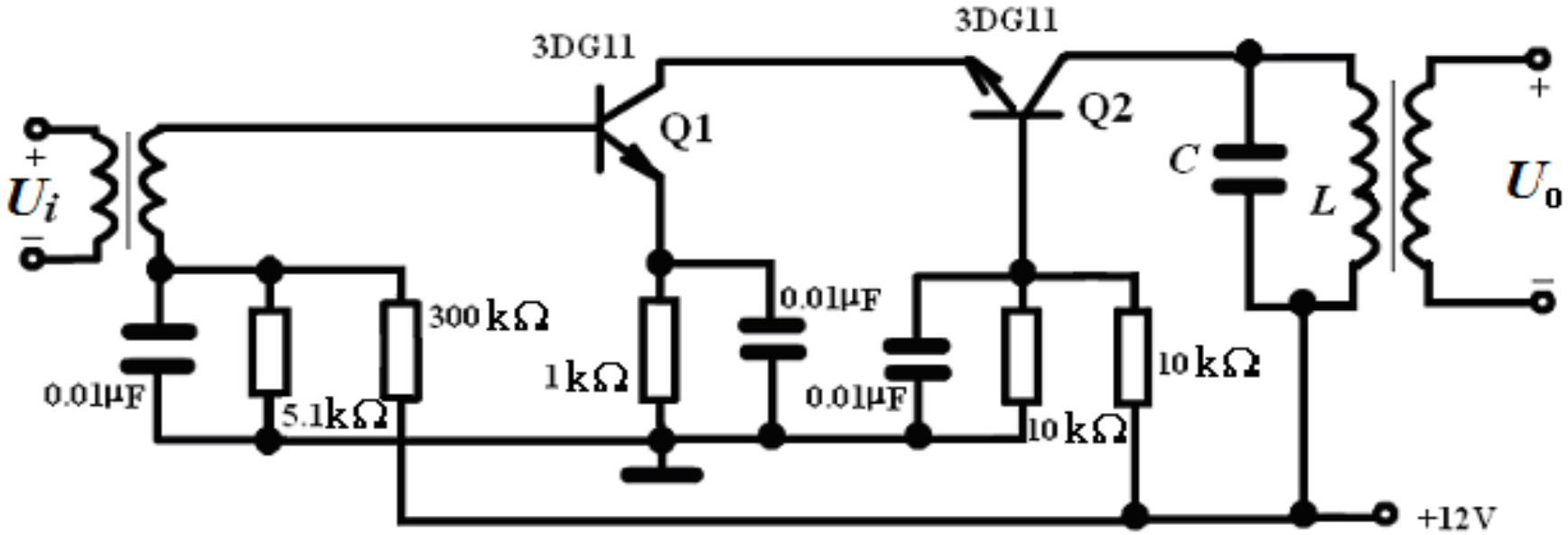
1.中和法： 在放大器的外部采用措施抵消内部反馈，从而实现放大器的单向化.



$$C_{N_1} \propto C_{b'c}$$

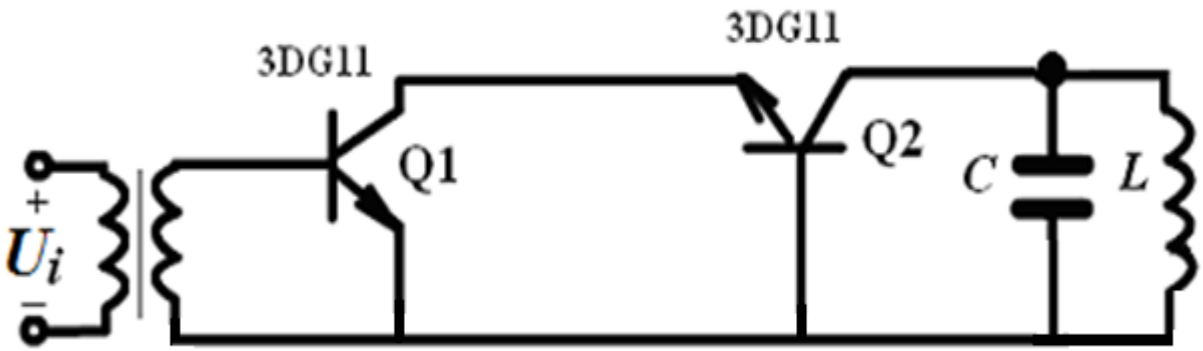
2.失配法：

以牺牲增益来换取放大器的稳定性.



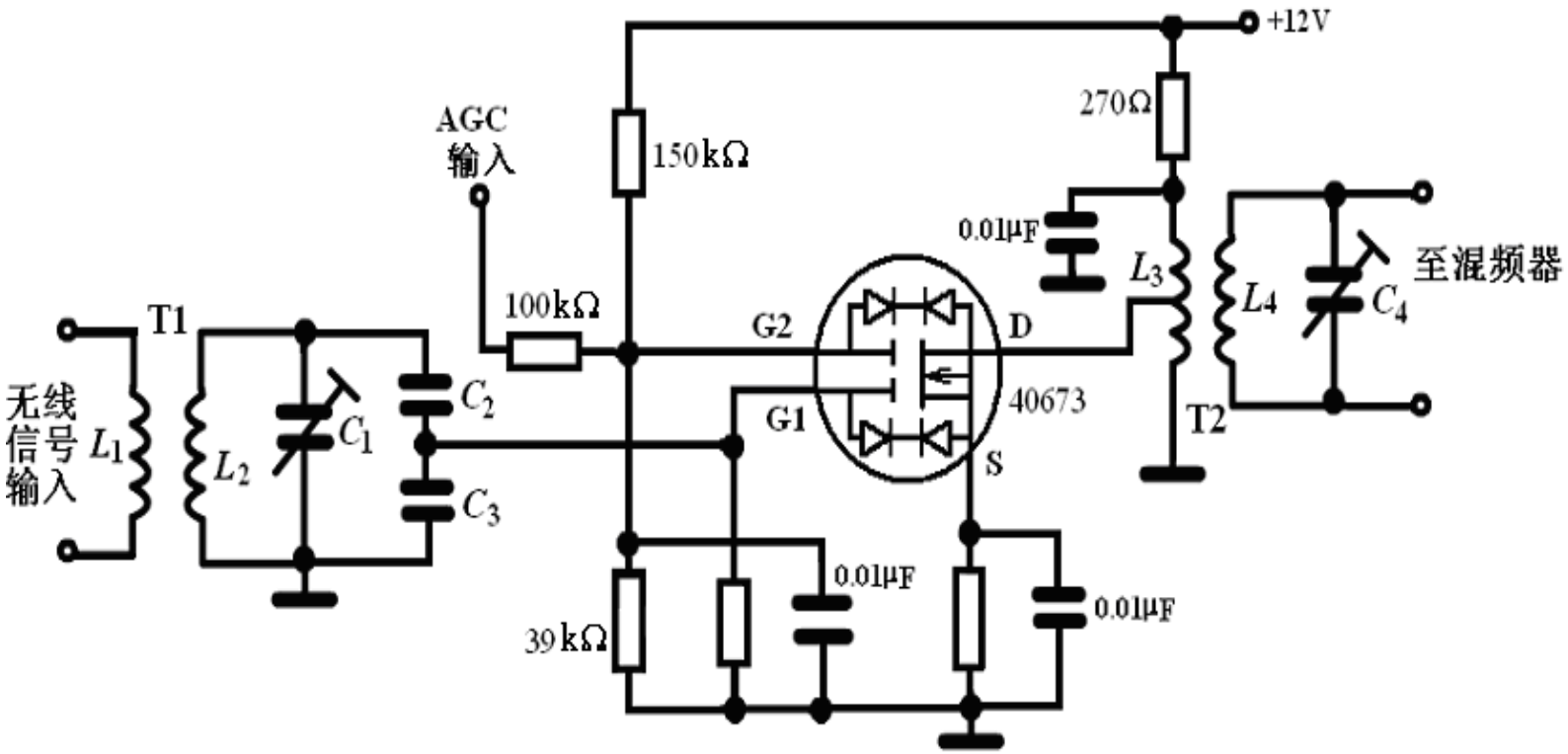
采用共射——共基放大回路减小内反馈的影响的效果较好

采用共射——共基放大回路减小内反馈的影响的效果较好



共基放大器的特点是输入阻抗小，输出阻抗大。
当它与共射放大器连接时，相当于共射的负载导纳很大。

3.选$c_{b'c}$小/ y_{re}小的晶体管,双栅场效应管



双栅极MOSFET RF放大器

例4

某中频放大线路如图所示，已知放大器的工作频率为$f_0 = 10.7\text{MHz}$，回路电容$C = 50\text{pF}$，中频变压器接入系数$p_1 = N_1/N = 0.35, p_2 = N_2/N = 0.03, Q_0 = 100$，晶体管Y参数(在工作频率上)如下：

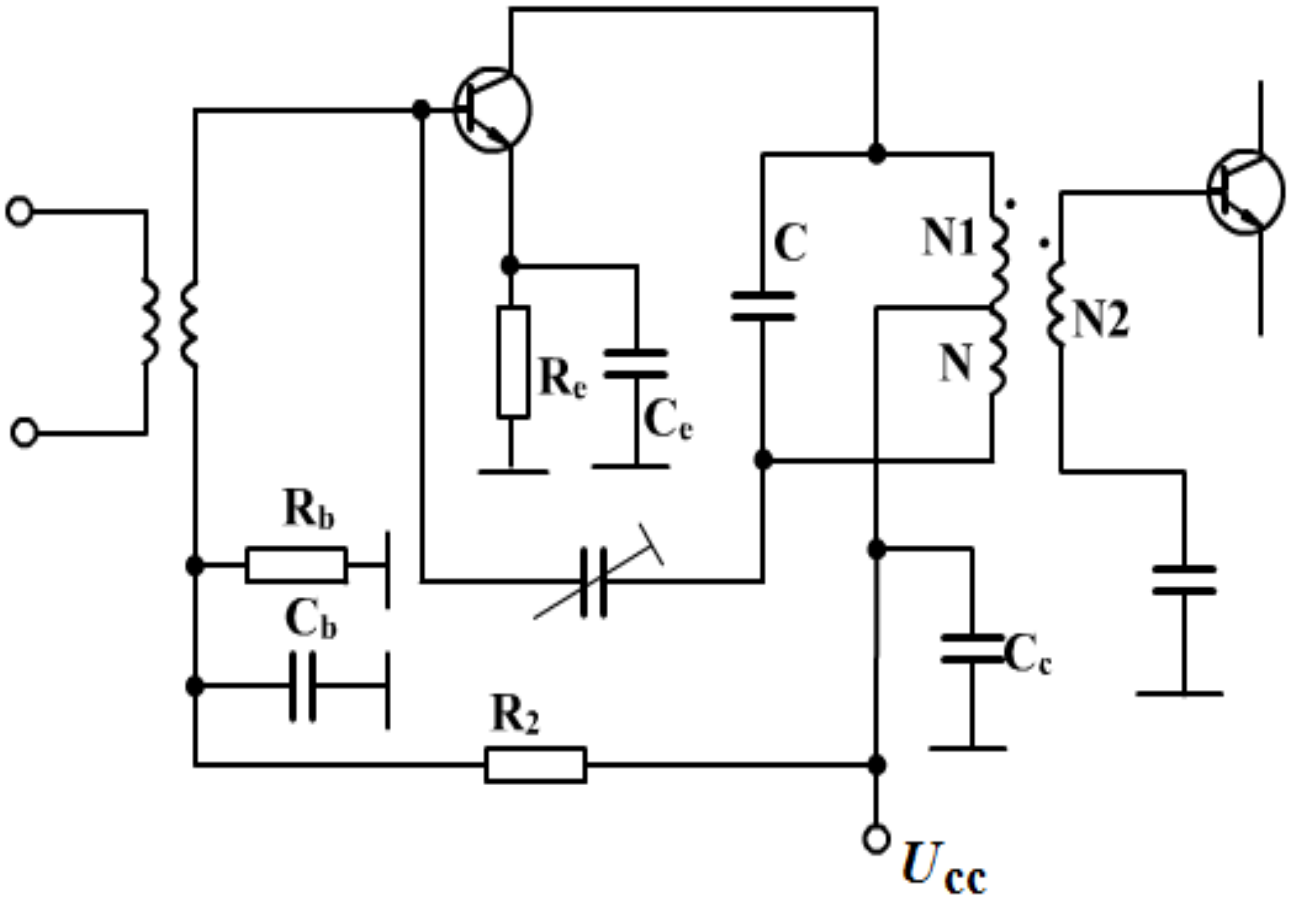
$$g_{ie} = 1.0\text{ms}, C_{ie} = 41\text{pF}, y_{re} = -j180\mu\text{s}.$$

$$y_{fe} = 40\text{ms}, C_{oe} = 4.3\text{pF}, g_{oe} = 45\mu\text{s}$$

设后级输入电导也为g_{ie}，求

(1)回路有载Q_L值和通频带BW

(2)放大器的谐振电压增益.


C
N1
N2
N
R_e
C_e
R_b
C_b
R_2
C_c
U_{cc}

要计算$Q_L \rightarrow BW = \dfrac{f_0}{Q_L}$，$\rightarrow Q_L$与有负载阻抗有关，

解

负载有三部分：回路本身负载R_p

$$R_p = \frac{Q_0}{\omega_0 C} = \frac{100}{2\pi \times 10.7 \times 10^6 \times 50 \times 10^{-12}} \approx 30\text{k}\Omega$$

晶体管输出阻抗：

$$R_{oe}' = \frac{1}{p_1^2 g_{oe}} = \frac{1}{0.35^2 \times 45 \times 10^{-6}} = 181\text{k}\Omega.$$

后级输入阻抗：

$$R_{ie}' = \frac{1}{p_2^2 g_{ie}} = \frac{1}{0.03^2 \times 1.0 \times 10^{-3}} = 1.1\text{M}\Omega$$

总阻抗: $R_T = R_p // R_{oe}' // R_{ie}' = 30k\Omega // 1.1M\Omega // 181k\Omega$

$$\approx 25k\Omega$$

回路总电容: $C_T = C + C_{ie}' + C_{oe}'$

$$= C + p_2^2 C_{ie} + p_1^2 C_{oe}$$

$$= 50 + 0.35^2 \times 4.3 + 0.03^2 \times 41 \approx 50.54pf$$

因此f_0基本不变,而

$$Q_L = \omega_0 C_T R_T = 2\pi \times 10.7 \times 10^6 \times 50.54 \times 10^{-12} \times 25 \times 10^3$$

$$= 84.9$$

$$BW = \frac{f_0}{Q_L} = \frac{10.7 \times 10^6}{84.9} = 126kHz$$

谐振时: $f = f_0, \xi = 0$

$$A_{V0} = -p_1 p_2 \cdot \frac{|y_{fe}|}{G_T} = -p_1 p_2 |y_{fe}| \cdot R_T$$

$$= -0.35 \times 0.03 \times 40 \times 10^{-3} \times 25 \times 10^{3}$$

$$= -10.5$$

$$20 \log_{10} A_{V0} = 20.4\text{dB}$$

§1.6 小信号集中选频放大器

高频小信号放大器分类

按频带分

窄带	选频回路，如LC谐振回路
宽带	电阻和高频变压器等宽带负载

按使用有源器件分

分立	调谐回路为主
集成	宽带负载集中选频放大器

1. 集中选频滤波器

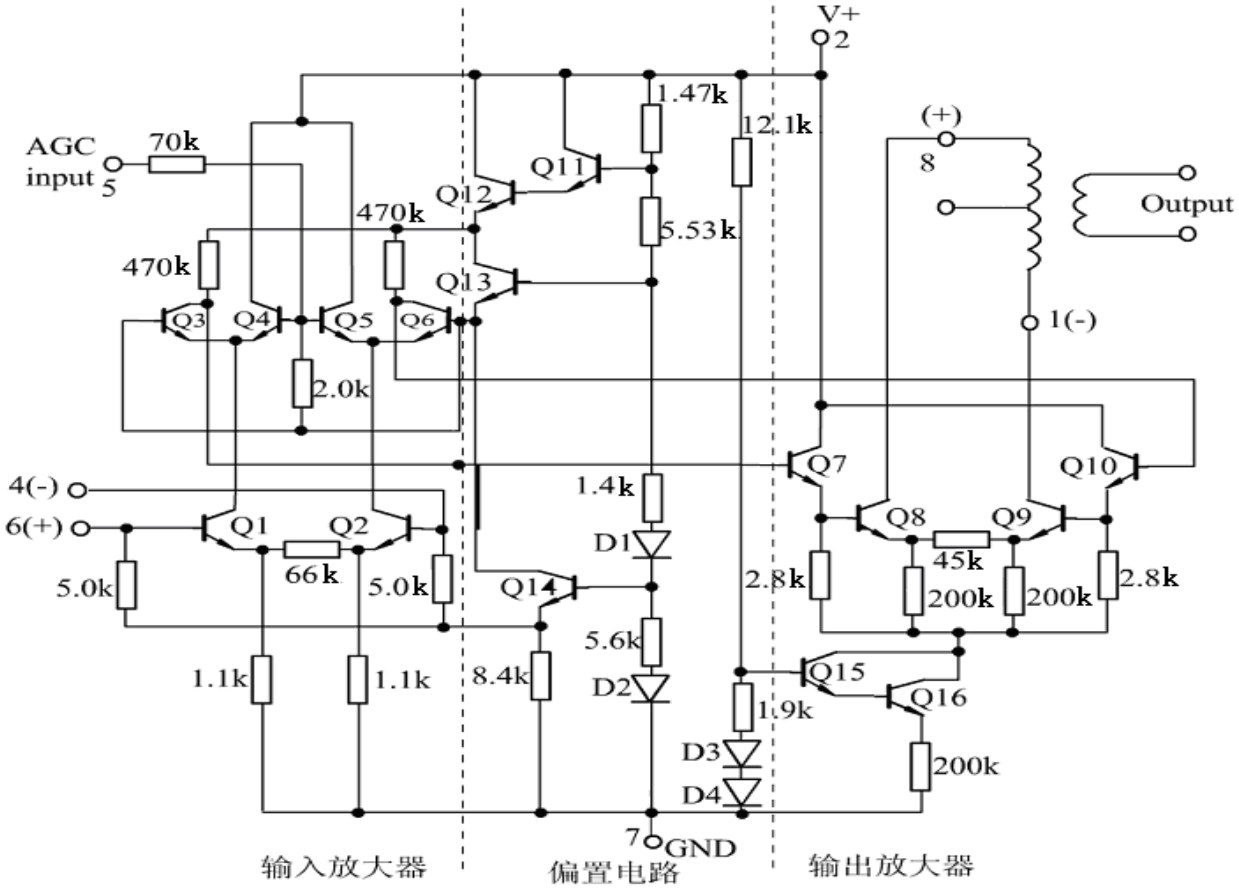
声表面波带通滤波器指标

中心频率MHz	10~1500
相对带宽 $\Delta f / f_0$	50%以上
最小带宽kHz	100
矩形系数	1.15
带外抑制dB	60以上
带内波动dB	±0.05
插入损耗	最大达25dB

- 压电陶瓷谐振器
- 声表面波滤波器

2. 多级宽带放大器

➢集成电路MC1350

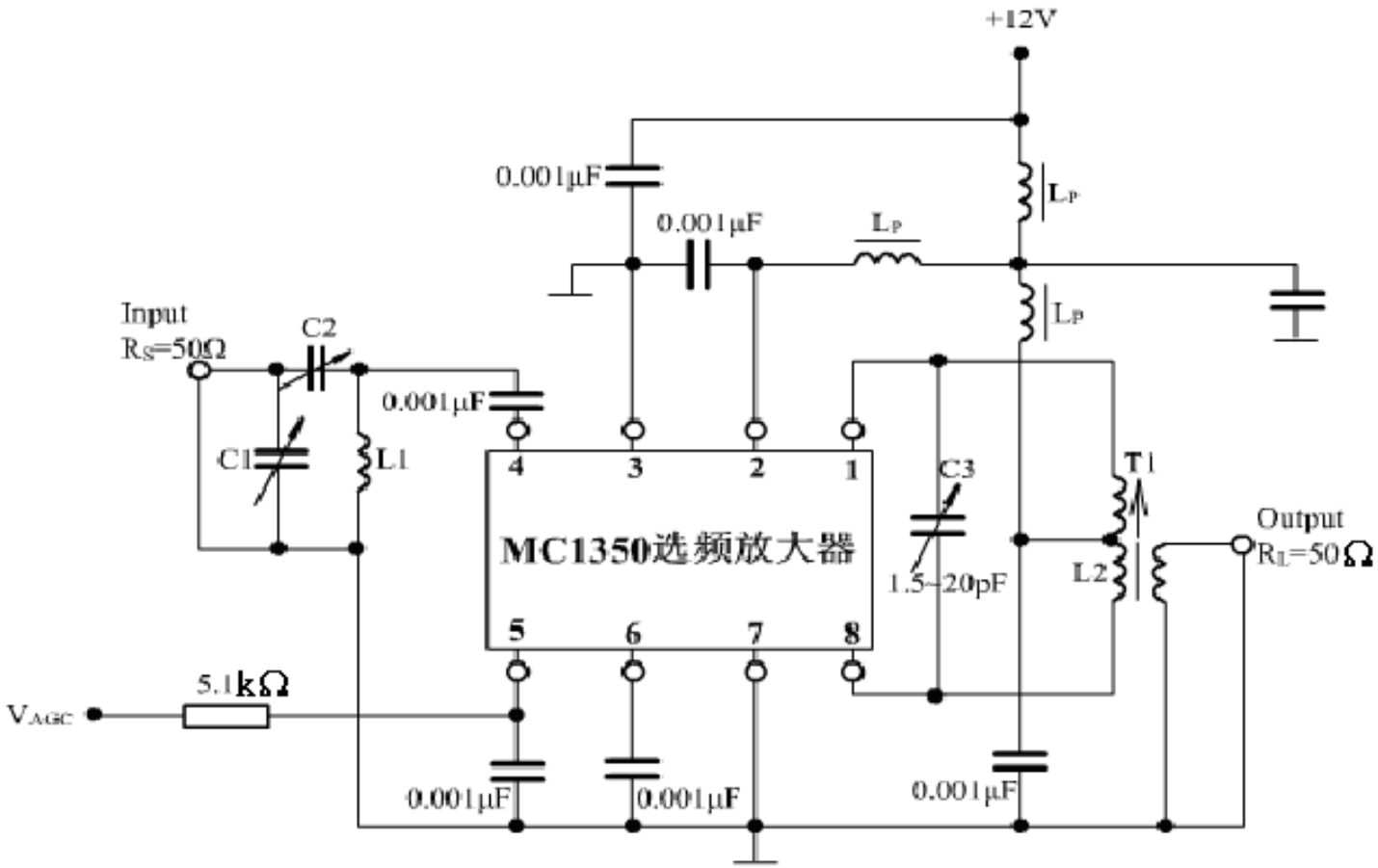


集成放大器MC1350
是用作RF/IF放大器的典型器件

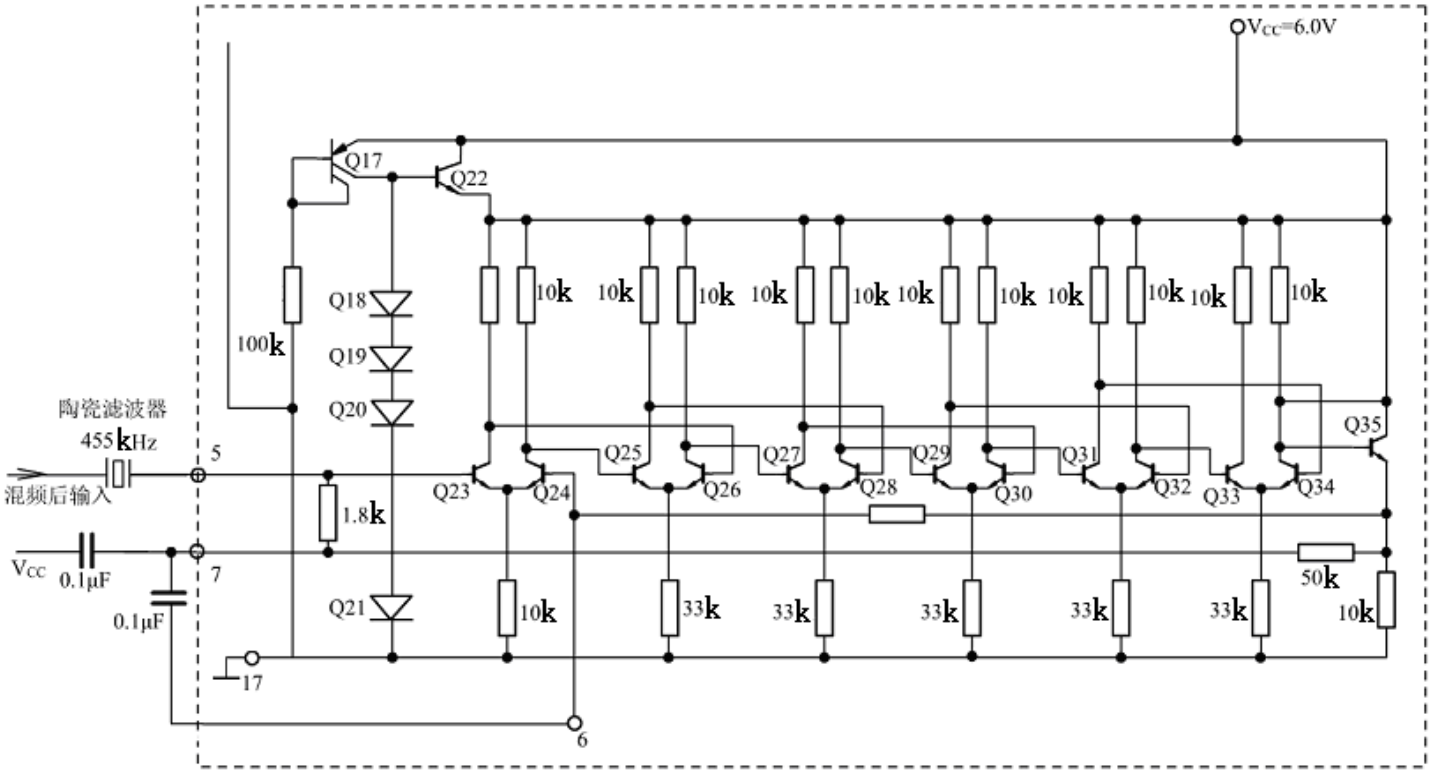
MC1350放大器主要参数

- **功率增益　　50dB**
- **AGC范围　+5V~+9V**
- **带宽　　58MHz和48MHz时 BW=4.5MHz**
  10.7MHz时BW=350kHz
  455kHz时BW=20kHz
- **噪声系数　≤8 dB**
- **电源电压　　18V消耗功率　　625mW**

1-3.exe


+12V
0.001μF
0.001μF
LP
LP
LP
Input
RS=50Ω
C2
0.001μF
C1
L1
4
3
2
1
MC1350选频放大器
5
6
7
8
C3
1.5~20pF
T1
L2
Output
RL=50Ω
5.1kΩ
VAGC
0.001μF
0.001μF
0.001μF

(a)

3 集成中频（IF）放大器

MC3359集成电路内部中频限幅放大级电路